A Minor Project Synopsis

On

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1. Introduction

Waste management is one of the most pressing urban challenges of the 21st century. Rapid urbanization has significantly increased the volume of solid waste generated daily, overwhelming conventional collection systems that operate on rigid, schedule-based routines regardless of actual bin fill levels. This leads to overflowing bins, health hazards, environmental degradation, and wasteful use of fuel and labour.

The Internet of Things (IoT) presents a transformative opportunity by enabling real-time monitoring of bin fill levels through low-cost sensors. Data is transmitted wirelessly to a cloud platform, where collection routes can be dynamically optimized — dispatching vehicles only when needed, reducing costs and improving sanitation.

This project — **IoT-Based Smart Waste Management System** (Project ID: BTP2CSE104) — proposes an end-to-end solution comprising smart bin hardware (ESP8266 + ultrasonic sensor), cloud connectivity via MQTT, and a React.js web dashboard with automated alerts. It is developed as a Minor Project for B.Tech (CSE), Session 2025-26, at IILM University, Greater Noida, under the guidance of Mr. Narinder Verma.

Layer	Component	Function
Hardware	ESP8266 + HC-SR04	Measure bin fill level continuously
Communication	MQTT over Wi-Fi	Transmit sensor data to cloud
Cloud	Firebase / Node.js	Store data & trigger alerts
Application	React.js Dashboard	Visualize live bin status on map

2. Problem Statement

Conventional waste collection is schedule-based and reactive, leading to multiple critical inefficiencies in urban sanitation management:

- Bins frequently overflow before scheduled pickup, creating health hazards and foul odour in public areas.
- Collection vehicles travel fixed routes regardless of whether bins are full or empty, wasting fuel and manpower.
- Municipal authorities have no real-time visibility into bin fill levels across the city.
- High operational costs due to redundant trips, excess labour deployment, and poor resource planning.
- Overflowing waste creates breeding grounds for disease vectors (mosquitoes, rodents), impacting public health.
- Environmental pollution from uncollected waste seeping into soil and water bodies.

An intelligent, automated system providing real-time bin monitoring, timely collection alerts, and dynamic route optimization is urgently needed to address these challenges.

Aspect	Current Problem	Proposed Solution
Collection Timing	Fixed schedule, ignores actual fill	Alert when bin exceeds 80% fill
Monitoring	No live data available	Cloud dashboard with live status
Route Planning	Predefined, inefficient routes	Dynamic routing based on fill data
Cost	High fuel & labour wastage	Est. 30-40% reduction in trips
Public Health	Frequent overflow, disease risk	Bins emptied before overflow

3. Literature Review

A review of existing literature reveals significant research on IoT-based waste management, establishing the technical and practical foundations for this project. Key studies are summarised below:

[1] Anagnostopoulos et al. (2015) – IEEE Trans. Sustainable Computing

Proposed a smart waste collection framework using ZigBee-based wireless sensor networks for bin fill monitoring. The study demonstrated that real-time data-driven collection can reduce unnecessary trips by up to 40%, forming the basis for dynamic route optimization adopted in this project.

[2] Gutierrez et al. (2015) – IEEE Access

Introduced 'SmartWaste', a cloud-connected bin monitoring system using ultrasonic sensors and GSM communication. The layered hardware-cloud-dashboard architecture directly inspired the system design of this project, particularly the threshold-based automated alert mechanism.

[3] Al-Masri et al. (2018) – IEEE Access

Conducted a comprehensive comparison of IoT messaging protocols (MQTT, CoAP, HTTP). The study concluded that MQTT offers the best balance of latency, bandwidth efficiency, and reliability for constrained IoT devices, directly justifying the protocol selection in this project.

[4] Pardini et al. (2020) – Journal of Sensor and Actuator Networks

Presented a systematic survey of IoT-based solid waste management solutions globally, identifying key gaps including absence of scalable open-API architectures — both addressed in this project's design.

[5] Lozano et al. (2016) – Procedia Computer Science

Implemented an Arduino and HC-SR04-based smart bin prototype with cloud integration, validating sensor accuracy of ±3 mm at ranges up to 400 cm — the same component adopted in this project's hardware design.

[6] Sharma & Singh (2021) – Int. Journal of Engineering Research

Developed an IoT waste bin system using NodeMCU ESP8266 and MQ-2 gas sensors with MQTT-over-Wi-Fi communication. Their threshold alert logic and firmware architecture directly informed the embedded design of this project.

Summary of Literature Review

Ref.	Author(s)	Key Contribution	Relevance
[1]	Anagnostopoulos et al. (2015)	ZigBee WSN bin monitoring	Route optimization
[2]	Gutierrez et al. (2015)	SmartWaste cloud bin system	System architecture
[3]	Al-Masri et al. (2018)	MQTT vs CoAP vs HTTP	Protocol selection
[4]	Pardini et al. (2020)	IoT waste mgmt survey	Design scope

[5]	Lozano et al. (2016)	HC-SR04 prototype validation	Sensor hardware
[6]	Sharma & Singh (2021)	NodeMCU firmware design	Firmware & MQTT

3. Objectives

The specific objectives of the IoT-Based Smart Waste Management System are:

No.	Objective	Description
1	**Hardware Design**	Design IoT-enabled smart bins with ESP8266 NodeMCU + HC-SR04 ultrasonic sensor for continuous fill-level measurement.
2	**Wireless Transmission**	Establish reliable MQTT-based Wi-Fi communication between bins and cloud server with <2s latency.
3	**Cloud Integration**	Integrate with Firebase Realtime Database for persistent data storage, live updates, and historical analytics.
4	**Web Dashboard**	Develop a React.js dashboard displaying map-view of all bins with colour-coded fill indicators (Green/Yellow/Red).
5	**Alert System**	Implement automated email/SMS notifications when any bin exceeds the 80% fill threshold.
6	**Route Optimization**	Build a logic module to suggest optimized collection routes based on real-time fill data, minimizing travel distance.
7	**Performance Testing**	Validate system for sensor accuracy (±2 cm), data latency (<2s), system uptime (>99%), and battery life (>48 hrs).
8	**Scalability**	Design a modular, extensible architecture supporting additional bins, sensor types, and GIS platform integration.

4. Methodology

The project follows a five-phase structured development methodology. Each phase is sequential but iterative, with feedback loops between testing and earlier phases.

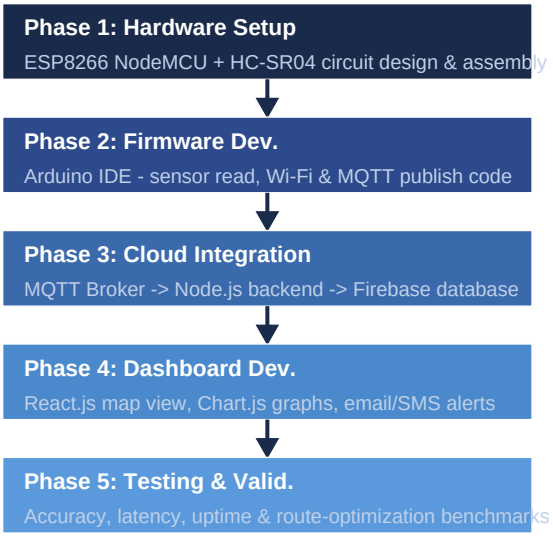

Figure 1: System Development Methodology Flowchart

Phase 1 – Hardware Setup: ESP8266 NodeMCU interfaced with HC-SR04 sensor; fill percentage computed from distance reading.

Phase 2 – Firmware Development: Arduino IDE (C/C++) code for periodic sensor reads, Wi-Fi connection, and MQTT message publishing.

Phase 3 – Cloud Integration: MQTT broker (HiveMQ) relays data to Node.js backend; stored in Firebase Realtime Database.

Phase 4 – Dashboard Development: React.js frontend with map view, fill indicators, Chart.js trend graphs, and automated alert dispatch.

Phase 5 – Testing & Validation: Sensor accuracy, latency, alert reliability, and performance benchmarking against KPIs.

5. Tools & Technologies

Hardware

Component	Details
Microcontroller	ESP8266 NodeMCU (Wi-Fi) / Arduino Uno (testing)
Sensor	HC-SR04 Ultrasonic (range 2-400 cm, ±3 mm accuracy)
Temp. Sensor	DHT11 – bin temperature & humidity monitoring
GSM Module	SIM800L – 2G fallback for Wi-Fi-absent areas
Power	5V USB / 3.7V Li-Po battery + solar charging option

Software & Cloud

Component	Details
Firmware IDE	Arduino IDE 2.x with ESP8266 board package (C/C++)
Backend	Node.js (Express) / Python (Flask) REST API
Frontend	React.js + Chart.js + Google Maps / Leaflet.js
Database	Firebase Realtime DB (live) + MongoDB Atlas (history)
Alerts	Twilio (SMS) / SendGrid (Email) notification services

Protocols & Tools

Component	Details
Protocol	MQTT v5.0 – lightweight publish-subscribe IoT protocol
MQTT Broker	HiveMQ Cloud (free tier) / Mosquitto (self-hosted)
Cloud IoT	AWS IoT Core (optional enterprise deployment)
Dev Tools	VS Code, Postman, Git & GitHub, Arduino Serial Monitor

6. Expected Outcomes

The following outcomes are expected upon successful completion of this project:

1 Functional Smart Bin Prototype
Fully operational IoT bin measuring fill level & temperature, transmitting data every 5 min via MQTT.

2 Real-Time Cloud Dashboard
React.js web app with map view, colour-coded fill indicators and live data updates.

3 Automated Alert System
Email/SMS alerts dispatched within 30 seconds of any bin crossing the 80% threshold.

4 Route Optimization
Dynamic route suggestion module identifying only bins needing collection, reducing trips by 30-40%.

5 Validated Performance
Sensor accuracy ±2 cm, data latency <2s, system uptime >99%, battery life >48 hrs.

6 Cost Savings Report
Comparative analysis demonstrating fuel, manpower, and operational cost reduction.

7 Scalable Architecture
Modular design supporting additional bins, sensor types, and GIS integration without redesign.

Fill Level	Range	Indicator	Action
Low	0% - 50%		No action needed
Medium	51% - 79%	**YELLOW**	Monitor closely
Critical	80% - 100%		Dispatch collection immediately

7. Applications

The system is versatile and applicable across diverse real-world domains:

Smart City Waste Mgmt.	Municipal corporations deploy system city-wide. Real-time bin data feeds a command centre for coordinated collection, aligning with Smart Cities Mission & Swachh Bharat Abhiyan.
University Campuses	IILM University and similar campuses can monitor bins across hostels, canteens & academic blocks, optimizing collection schedules and promoting cleanliness.
Commercial Complexes	Shopping malls generate high irregular waste volumes. Smart bins prevent overflow near food courts without disrupting visitor experience.
Healthcare Facilities	Hospitals track general and biohazardous waste bins; temperature sensors detect anomalies and ensure timely, regulation-compliant disposal.
Industrial Facilities	Factories monitor bulk or hazardous waste containers remotely, aligning collection with production cycles and reducing environmental violations.
Event Management	Festivals and public events deploy temporary smart bins for dynamic collection response during high-density waste generation periods.
Rural & Semi-Urban Areas	GSM-based variant (SIM800L) extends smart waste management to areas without Wi-Fi, ensuring equitable coverage across infrastructure gaps.
GIS & Smart City Platforms	Open API architecture enables integration with city-level GIS systems, providing unified waste logistics analytics on a single municipal platform.

8. Conclusion

The IoT-Based Smart Waste Management System addresses a critical real-world problem by integrating low-cost IoT hardware, MQTT-based wireless communication, cloud computing, and an interactive web dashboard into a unified, intelligent solution. By transitioning from reactive, schedule-based collection to proactive, data-driven management, the system offers measurable improvements in efficiency, cost savings, and public sanitation.

The system's core innovation — converting real-time sensor data into actionable collection intelligence — is expected to reduce unnecessary collection trips by 30-40%, lower fuel and labour costs, and eliminate bin overflow incidents. Automated alerts and dynamic routing ensure timely response without manual intervention.

From a technical and academic standpoint, this project provides the team with hands-on experience across embedded systems (ESP8266, Arduino), IoT protocols (MQTT), full-stack development (React.js, Node.js, Firebase), and cloud integration — directly aligned with industry requirements and the B.Tech (CSE) curriculum at IILM University.

The modular, open-API architecture ensures straightforward extension in future work: AI/ML-based fill prediction, gas and weight sensors for waste classification, mobile app for field teams, and full GIS integration with smart city platforms.

In conclusion, this project demonstrates a meaningful, scalable application of engineering knowledge to improve urban living standards. It aligns directly with the national goals of Swachh Bharat Abhiyan and the Smart Cities Mission, and serves as a strong foundation for future smart city waste logistics solutions.

References

[1] Anagnostopoulos, T. et al. (2015). Challenges and Opportunities of Waste Management in IoT-Enabled Smart Cities. IEEE Trans. Sustainable Computing.

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[6] ESP8266 NodeMCU Documentation: https://nodemcu.readthedocs.io

[7] Firebase Realtime Database Docs: https://firebase.google.com/docs/database

[8] MQTT Protocol v5.0 Specification: https://mqtt.org/mqtt-specification

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